

VENUGOPAL INDRAJITH HEMASHREE

Chennai, Tamil Nadu, India | +91-9444668821 | shreejith2005@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Vellore Institute of Technology, Chennai

B.Tech, Computer Science Engineering — Cyber Physical Systems | CGPA: 7.81

Aug 2023 – Present

Maharishi International Residential School, Kanchipuram

Class 12 (CBSE): 79% | Class 10 (CBSE): 89%

PATENT

IoT-Enabled Smart Walker with Sensor-Fusion Fall Risk Prediction and Adaptive Mobility Assistance

Co-Inventor | Status: Filing in Progress (Inventive disclosure filed with IIPRD, Khurana & Khurana) | Faculty Guide: Dr. Subitha D

Arduino Uno · Dual MPU6050 · Dual HX711 · Dual HC-SR04 | [GitHub](#)

- ▶ Co-invented an Arduino Uno-based smart walker combining dual MPU6050 IMUs, dual HX711 load cells, and dual HC-SR04 ultrasonic sensors for real-time bilateral fall detection, weight-asymmetry monitoring, and obstacle avoidance, running fully standalone with no cloud dependency at an estimated hardware cost under USD 30.
- ▶ Designed a sensor-fusion Fall Risk Prediction engine and a Dynamic Stability Reserve metric that combine tilt, weight imbalance, and obstacle proximity to flag deteriorating stability before a fall occurs, alongside near-fall detection and a Gait Symmetry Index for rehabilitation tracking.

PROJECTS

AI-Based Dental Malocclusion Detection and Teeth Alignment Analysis

Jul 2025 – Sep 2025

Python, YOLOv9, Grad-CAM, Gradio, OpenCV, Deep Learning, Jupyter Notebook | [GitHub](#)

- ▶ Built a YOLOv9-based detection pipeline for automated dental malocclusion and teeth-alignment analysis from X-ray images.
- ▶ Applied Grad-CAM to visualize the regions driving each prediction, supporting clinical interpretability and orthodontic decision support.
- ▶ Deployed an interactive Gradio interface for real-time image upload, prediction, and result visualization.

Reinforcement Learning-Driven Autonomous Rover for Hazard Detection and Environmental Exploration

2026

Python · Raspberry Pi 5 · BreezySLAM · YOLOv8n / YOLOv9t · A* · Reinforcement Learning · Flask

- ▶ Engineered an autonomous ground rover that performs systematic area sweeps for disaster-response reconnaissance, running LIDAR-based SLAM (BreezySLAM) on a self-upgraded Raspberry Pi 5 to build a real-time map of the environment overlaid with a risk heatmap of detected hazard zones.
- ▶ Modified the YOLOv8n architecture directly — adding a multi-task risk-score prediction head and reweighting the loss function to prioritize human detection — for real-time victim detection, running alongside a separate camera-based YOLOv9t model and an MQ-2 gas sensor for on-board fire/smoke/gas hazard detection.
- ▶ Designed a modified A* search over the live SLAM map to compute the safest and shortest evacuation path out of the hazard zone for detected humans, paired with generated voice commands to guide them to safety in real time.
- ▶ Implemented reinforcement-learning-based navigation so the rover adapts its sweep path around newly detected hazards in real time, streaming live position, sensor readings, and detections to a Flask telemetry dashboard for human oversight.
- ▶ Fused HC-SR04 ultrasonic sensors, Pi Camera, and an MPU9250 IMU across both autonomous and manual operation modes, keeping navigation robust when individual sensors are degraded.

Smart Grocer — AI-Powered Produce Spoilage Detection & Inventory System

In Development

Python, YOLO, Computer Vision, Sensor Fusion, ESP32/Raspberry Pi, Firebase, Flutter | [GitHub](#)

- ▶ Architecting an AI-powered multimodal system for fresh-produce spoilage detection, individual-item localization, and expiry-aware retail inventory management, combining computer-vision-based produce classification with gas/VOC sensor fusion.
- ▶ Designing a YOLO-based pipeline to detect and classify individual produce items by freshness stage (fresh / ripening / at-risk / spoiled), with a rack-level anomaly-detection baseline (rolling statistics, z-score) to flag abnormal spoilage trajectories.
- ▶ Developing a sensor-fusion freshness engine that combines visual, gas, and environmental confidence scores into a unified spoilage-risk score, alongside a localized gas-sampling module that targets VOC measurement to camera-detected item coordinates.
- ▶ Planning barcode/OCR-based expiry tracking and a Firebase + Flutter mobile app for real-time rack monitoring, inventory, and waste analytics.

INTERNSHIP

Research Intern — VIT Chennai, under Prof. Pavithra

May 2025 – Jul 2025 (1 month)

Human Pose Estimation using Heatmaps (MMPose Validation) | MMPose, OpenCV, Kinovea, Matplotlib, Python

- ▶ Implemented human joint detection using heatmap-based pose estimation models with MMPose, visualizing and analyzing joint-confidence heatmaps for keypoint localization.
- ▶ Validated pose-estimation output against ground truth using Kinovea motion-analysis software to sanity-check keypoint accuracy under faculty supervision.

TECHNICAL SKILLS

- ▶ **Languages:** Java, Python
- ▶ **Web Development:** HTML, CSS, JavaScript
- ▶ **ML / Computer Vision:** YOLOv8/v9, MMPose, OpenCV, Grad-CAM, TensorFlow/PyTorch, Heatmap-based Keypoint Detection
- ▶ **Cybersecurity:** Intrusion Detection, Threat Modeling, Risk Analysis
- ▶ **Embedded / IoT:** Arduino, Sensor Fusion
- ▶ **Databases & Tools:** MySQL, Git, VS Code, Jupyter Notebook, Google Colab, Gradio
- ▶ **Spoken Languages:** Tamil (Native), English (Fluent), Hindi (Intermediate), Japanese (Basic)

ACHIEVEMENTS & LEADERSHIP

- ▶ **IBM Generative AI Certification** — “Gen AI Using IBM Watsonx,” IBM Career Education Program, via IBM Developer Skills Network (Jun 2025).
- ▶ **NUS Introduction to Industry 4.0 Programme, National University of Singapore** (Jun 2025) — completed via the School of Continuing and Lifelong Education (SCALE) & College of Design and Engineering; team member on a final project on digital transformation using the Business Model Canvas.
- ▶ **SuperAI Conference 2025, Marina Bay Sands, Singapore** (Jun 2025) — attended Asia's largest AI conference, featuring 7,000+ innovators and 1,000+ AI companies.
- ▶ **Team Leader, CodeYGen Club** — led a student team through coding and design-based projects, coordinating development and presentations.
- ▶ **Design Lead, Fitness Club** — designed promotional creatives, posters, and digital content to support fitness events and member engagement.